



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:



Course Name: B.tech
Subject Name: DS
Subject Code: 25CS101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	Draw Memory hierarchy	CO1	K2	2.1.1
(b)	Example of syntax and semantic error	CO1	K2	2.1.1
(c)	Define the characteristics of an Algorithm	CO1	K2	2.1.2
(d)	Find the value of $z = (10 * 0)$ in the expression	CO2	K2	2.1.1
(e)	Find the value of $z = ++x + ++x$, where $x = 10$	CO2	K2	2.1.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Difference between compiler and interpreter	CO1	K1	2.1.2
(b)	Draw flow chart for temperature conversion from Fahrenheit to Celsius(hint: $-f - 32 = 9c/5$)	CO1	K1	2.1.1
(c)	Difference between Multiprogramming and Multitasking	CO1	K1	1.4.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Draw block Diagram of computer and explain the components	CO1	K2	1.4.1
(b)	Why Operating System also known as resource manager?	CO1	K2	1.4.1
4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Write a C++ program to input a number and check whether it is even or odd using ternary operator.	CO2	K3	1.4.1
(b)	Write a C++ program to input the marks of three subjects and calculate the average marks. Display the result.	CO2	K3	1.4.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

HOD



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-II (Session: 2025-26)

Roll No. [REDACTED]

Course Name: B.Tech.
Subject Name: Data Structure
Subject Code: 25CS101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	Write the condition for accessing the diagonal elements of square matrix of order N.	CO3	K1	
(b)	Write a C program to print all natural numbers from 10 to 100.	CO3	K1	
(c)	Define an 2-D Array.	CO3	K1	
(d)	Compare if-else-if ladder and switch case.	CO2	K1	
(e)	Write the syntax for WHILE loop.	CO2	K1	
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Write a C program to input a number and check whether it is palindrome or not.	CO3	K2	
(b)	Differentiate between Break and Continue with the help of a program.	CO3	K2	
(c)	Differentiate between while and do while loop.	CO3	K2	
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Write a program in C++ to print the given pattern: <pre>***** ***** ***** **** *** ** *</pre>	CO3	K3	
(b)	Write a program in C++ language to multiply 2 matrices of order NxN each and print the resultant matrix.	CO3	K3	
4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Explain the use of default in switch statement. Write a program that takes two operands and one operator (+, -, *, /) from the user and perform the operation and prints the result by using switch statement.	CO2	K3	

(b)	Write a program in C++ to print grades as per following criteria for obtained percentage of marks M out of 100:		CO2	K3														
	<table><tr><th>Obtained Percent Marks (M)</th><th>Grade</th></tr><tr><td>90 <M≤ 100</td><td>A+</td></tr><tr><td>80 <M≤ 90</td><td>A</td></tr><tr><td>70 <M≤ 80</td><td>B+</td></tr><tr><td>60 <M≤ 70</td><td>B</td></tr><tr><td>50 <M≤ 60</td><td>C</td></tr><tr><td>M≤ 50</td><td>F</td></tr></table>				Obtained Percent Marks (M)	Grade	90 <M≤ 100	A+	80 <M≤ 90	A	70 <M≤ 80	B+	60 <M≤ 70	B	50 <M≤ 60	C	M≤ 50	F
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	90 <M≤ 100	A+																
	80 <M≤ 90	A																
	70 <M≤ 80	B+																
	60 <M≤ 70	B																
	50 <M≤ 60	C																
M≤ 50	F																	

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Course Name: B.Tech
Subject Name: Data Structure
Subject Code:25CS101

Time: 90 Min
Semester: I

Note:

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SECTION-A (Knowledge Level: K2 & K3)

Q. No.	Question	CO	KL	PI
Attempt any one part from each question		(3 Marks x 4 Questions = 12 Marks)		
1.(a)	What do you mean by Modular Programming ? Explain all types of functions with the help of an example.	CO4	K2	
1.(b)	Write a Program in C++ to find out the factorial of a given number using Recursion.	CO4	K2	
2.(a)	Write a Program in C++ to find out the area and circumference of a circle using functions with parameters.	CO4	K2	
2.(b)	Write a Program in C++ to find out whether a given number is prime or not using function.	CO4	K3	
3.(a)	What do you mean by Class and Object ? Explain all the properties of Object Oriented Programming.	CO5	K2	
3.(b)	What is String ? Write a Program in C++ to print ASCII value of all the characters of a string.	CO5	K2	
4.(a)	Write a Program in C++ to Concatenates two strings and print the length of a concatenated string.	CO5	K3	
4.(b)	Write a Program in C++ to replace first 'a' and last 'a' of given string with 'A'.	CO5	K3	

SECTION-B (Knowledge Level: K3)

Attempt any one part from each question		(4 Marks x 2 Questions = 8 Marks)		
5.(a)	What is Recursion ? Write a Program in C++ to print Fibonacci series using Recursion.	CO4	K3	
5.(b)	Explain the difference between call by value and call by reference with the help of an example.	CO4	K3	
6.(a)	Write a Program in C++ to find out whether a given string is Palindrome or not without using in-built function.	CO5	K3	
6.(b)	Write a Program in C++ to print all the vowels present in a string and also count no. of vowels	CO5	K3	

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create



Course Name: B.Tech
Subject Code: 25CS101

Admission No.:

2 [Redacted] 2
Max. Marks: 20

Semester: I

Note:

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SECTION-A (Knowledge Level: K2 & K3)

Q. No.	Question	CO	KL	PI
Attempt any one part from each question		(3 Marks x 4 Questions = 12 Marks)		
1.(a)	Define a function in C++. Explain the different type of functions based on input and return type.	CO4	K2	
1.(b)	Explain the concept of function declaration, definition and function call with an example.	CO4	K2	
2.(a)	Write a C++ function to find ab using recursion. where a,b are user input values.	CO4	K2	
2.(b)	Write a C++ function to find the sum of digits of a number using recursion.	CO4	K3	
3.(a)	What is a string in C++? Mention any two ways of string representation. Differentiate between strcpy() and strcat() functions.	CO5	K2	
3.(b)	What is OOPS? Explain the basic structure of a class in C++. Write a program to create a class and object.	CO5	K2	
4.(a)	Explain five commonly used string functions in C++ with the help of program.	CO5	K3	
4.(b)	Given two strings S1 and S2, concatenate them by inserting a space between them. Write a C++ program to print the concatenated string and its length.	CO5	K3	

SECTION-B (Knowledge Level: K3)

Attempt any one part from each question		(4 Marks x 2 Questions = 8 Marks)		
5.(a)	Write a C++ program to swap two numbers using call by value and call by reference. Also differentiate both concepts.	CO4	K3	
5.(b)	Define Recursion. Write a C++ program to print the multiplication of all the digits of a number using recursion.	CO4	K3	
6.(a)	Write a C++ Program for the given problem statement:- Given a string S, check whether it is a palindrome and its length is even. If both conditions are satisfied, print "YES" otherwise print "NO".	CO5	K3	
6.(b)	Write a C++ Program for the given problem statement:- Given a string S, determine whether it contains at least one digit and at least one alphabet. If both conditions are satisfied, print "YES" otherwise print "NO".	CO5	K3	

CO Course Outcomes mapped with respective question
KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create